FFGFT Analytics: Mathematical Foundations

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Contents

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FFGFT Analytics: Mathematical Foundations

Fractal Feedback and Matrix Translation in the 4D Phase Torus

Integration of TFLN Hardware, B-Meson Anomalies
and Sub-Planckian Frequency Metric

Info

Abstract. Fundamental Fractal Geometric Field Theory (FFGFT) describes space-time as a hierarchical system of fractal foldings, driven by a fundamental frequency ξ in the sub-Planckian regime. This document lays out the complete mathematical foundations: the exact definition of the fundamental tick t_0, the feedback condition in the closed 4D phase torus (which must not be confused with a 3D torus plus time continuum), the translation of the 4D vector space into a multidimensional matrix space, and the photonic realisation through TFLN hardware.

Cross-references: Doc. 180 (Lagrangian derivation) · Doc. 182 (Universal scale) · Doc. 183–184 (Precision barrier & RSA) · GitHub: jpascher/T0-Time-Mass-Duality · Zenodo: DOI 10.5281/zenodo.20474821

Contents

.1 Introduction: FFGFT

Fundamental Fractal Geometric Field Theory (FFGFT) breaks with the classical understanding of continuous space. It postulates a discrete fractal geometry energised by a fundamental oscillation. Matter is not an entity *in* space but a local condensation of the fractal field geometry, stabilised by resonant feedback.

.2 The Fundamental Frequency ξ and the Sub-Planckian Time Scale t_0

A core pillar of FFGFT is the existence of a sub-Planckian fundamental tick that clocks all of physical reality.

Definition of the Fundamental Tick

The fundamental time unit t_0 is defined as:

$$t_0 = \frac{t_P}{7500} \approx \frac{5.391 \times 10^{-44}\,\text{s}}{7500} \approx 7.188 \times 10^{-48}\,\text{s}, \tag{1}$$

where $t_P = \sqrt{\hbar G/c^5} \approx 5.391 \times 10^{-44}$ s is the Planck time (CODATA 2022). The resulting **fundamental frequency** ξ reads:

$$\xi = \frac{1}{t_0} = \frac{7500}{t_P} \approx 1.391 \times 10^{47}\,\text{Hz}. \tag{2}$$

Kernpunkt

Origin of the factor 7500. The factor 7500 arises from $\xi = 4/30000$ in natural units ($\xi = 2/(15000)$ in the original notation). The fractal dimension $D_f = 3 - \xi$ and the Z3 symmetry of the three-generation structure *enforce* this value geometrically — it is not a free parameter.

Frequency as Geometric Operator

In FFGFT, frequency is not understood as mere temporal rate but as a geometric operator acting on the fractal levels:

$$\hat{\xi}\,\Psi(x^\mu) = \xi \cdot \mathscr{F}(\{n_k\}) \cdot \Psi(x^\mu). \tag{3}$$

Here $\mathscr{F}(\{n_k\})$ is a function of the fractal scaling indices n_k, which follow the Fibonacci sequence.

.3 The 4D Phase Torus: Distinction from the 3D Torus

Warning: Not to be Confused

Hinweis

The 4D torus of FFGFT is **not** a 3D object moving through time. The time dimension is integrated into the torus topology as a discretised, geometric dimension. There is no external time continuum into which the torus would be embedded.

Formal Definition of the 4D Phase Torus

The fundamental spacetime resonator is the 4D phase torus $\mathscr{T}^4$, defined by four cyclic coordinates $(\theta_1, \theta_2, \theta_3, \theta_4)$ with:

$$\theta_i \in [0, 2\pi), \quad i = 1, 2, 3, 4. \tag{4}$$

The metric on $\mathcal{T}^4$ is determined by the fundamental frequency ξ:

$$ds^2 = \xi^{-2}\big(d\theta_1^2 + d\theta_2^2 + d\theta_3^2 + d\theta_4^2\big). \tag{5}$$

The fourth dimension θ_4 represents the phase shift of the fundamental oscillation. Macroscopic time t emerges by accumulation:

$$t = N \cdot t_0, \quad N = \frac{1}{2\pi}\oint d\theta_4. \tag{6}$$

.4 Fractal Feedback and Locking of the Fundamental Frequency

The Feedback Condition

For a field configuration Ψ to exist stably, the wave function must interfere constructively with itself after a complete circuit. The phase condition reads:

$$\oint_{\mathcal{T}^4} \nabla_\mu \phi \, dx^\mu = 2\pi \cdot \mathcal{N}, \tag{7}$$

where $\mathcal{N}$ is an integer from the fractal hierarchy. Due to the fractal structure, $\mathcal{N}$ can only take discrete values:

$$\mathcal{N} \in \{F_k \mid k \in \mathbb{N}\}, \quad F_{k+1} = F_k + F_{k-1}. \tag{8}$$

The Lock-In Mechanism (Phase Locking)

Fractal feedback enforces a rigid coupling between the phases of the individual fractal levels:

$$\phi_{k+1} = \phi_k \cdot \phi^{-1}, \quad \phi = \tfrac{1+\sqrt{5}}{2} \approx 1.6180\ldots \tag{9}$$

Only frequencies that exhibit exactly an integer or Fibonacci-sequential harmony with ξ remain stable:

$$\xi_n = n \cdot \xi \quad \text{or} \quad \xi_n = \phi^{-k} \cdot \xi. \tag{10}$$

Every deviation leads to destructive interference — which explains the short lifetimes of high-energy particles.

Stability Condition for Matter

From the lock-in follows the quantisation condition for stable particles:

$$\oint_{\mathcal{T}^4} \Psi_{\mathsf{FFGFT}}(\phi)\, d^4\theta = \xi \cdot \sum_{k=0}^{\infty} \phi^{-k} \cdot \mathbf{M}_k, \tag{11}$$

where $\mathbf{M}_k$ are the projective matrices of the individual fractal levels.

.5 From the 4D Vector Space to the Multidimensional Matrix Space

The General Matrix Translation

A state in the 4D vector space $V = (x, y, z, ct)$ is represented by a Hermitian matrix $\mathbf{H}$:

$$\mathcal{T} : \mathbb{R}^4 \to \mathrm{Mat}(N, \mathbb{C}), \quad V \mapsto \mathbf{H}(V). \tag{12}$$

The explicit form of the translation reads:

$$\mathbf{H}(x, y, z, ct) = \sum_{k=0}^{\infty} \alpha^k \cdot \begin{pmatrix} ct_k + z_k & x_k - iy_k \\ x_k + iy_k & ct_k - z_k \end{pmatrix} \otimes {}_k, \tag{13}$$

with:

- $\alpha = \phi^{-1}$: fractal coupling constant;
- (x_k, y_k, z_k, ct_k): coordinates on the k-th fractal level;
- ${}_k$: generators of the spacetime geometry (generalisation of the Dirac matrices).

Fractal Self-Similarity

The coordinates on different fractal levels are connected by scaling factors:

$$(x_{k+1}, y_{k+1}, z_{k+1}, ct_{k+1}) = \phi^{-1} \cdot (x_k, y_k, z_k, ct_k). \tag{14}$$

.6 Matrix Multiplication as Physical 4D Convolution

Conventional vs. Fractal Matrix Multiplication

In conventional computer science:

$$C_{ij} = \sum_k A_{ik} B_{kj}. \tag{15}$$

In FFGFT, this process is interpreted as physical interference of two 4D torus states.

The 4D Torus Convolution

Matrix multiplication corresponds to the convolution of two wave functions in 4D phase space:

$$M_{ij} = \int_{\mathcal{T}^4} \Psi_A^{(4D)}(\theta) \cdot \Psi_B^{(4D)}(\theta)\, d^4\theta. \tag{16}$$

In the language of FFGFT matrices:

$$\mathbf{C} = \mathbf{A} \star \mathbf{B} = \lim_{N \to \infty} \frac{1}{N} \sum_{n=1}^{N} \mathbf{A}_n \otimes \mathbf{B}_n, \tag{17}$$

where the fractal tensor product is defined by:

$$(\mathbf{A} \otimes \mathbf{B})_{ij}^{kl} = A_{ij} \cdot B_{kl} \cdot e^{i\theta_{ijkl}}. \tag{18}$$

Physical Interpretation

Photonic matrix multiplication in the TFLN crystal is not “computation” in the conventional sense, but the physical realisation of this 4D phase interference. The result emerges instantaneously through constructive and destructive interference of light waves in the nonlinear medium.

.7 Hardware Transduction: TFLN Photonics

Structural Equivalence Despite Vastly Different Scales

Important Clarification

The TFLN crystal operates on scales many orders of magnitude above the fundamental structure of particles or individual atoms. The crystal lattice is macroscopic — there is no direct scale comparison with ξ or t_0.
That is precisely the point: FFGFT postulates *fractal self-similarity* — the same mathematical structure appears on every scale, independent of the concrete order of magnitude.

The decisive observation is structural. In FFGFT, for the translation of a 4D vector state into matrix space (Section 5):

$$\mathcal{T} : \mathbb{R}^4 \to \mathrm{Mat}(N, \mathbb{C}), \quad V \mapsto \mathbf{H}(V), \tag{19}$$

and for the physical interaction of two states:

$$M_{ij} = \int_{\mathcal{T}^4} \Psi_A^{(4D)}(\theta) \cdot \Psi_B^{(4D)}(\theta)\, d^4\theta. \tag{20}$$

In TFLN waveguides, the nonlinear light propagation realises exactly this structure:

$$P_{\text{out}}(t) = \eta \cdot \int \chi^{(2)}(\omega) \cdot E_{\text{in}}^{(A)}(\omega) \cdot E_{\text{in}}^{(B)}(\omega)\, d\omega. \tag{21}$$

The nonlinear susceptibility $\chi^{(2)}$ takes the role of the fractal coupling constant $\alpha = \phi^{-1}$: it weights how two input states combine into an output state — mathematically identical to the 4D torus convolution, physically realised in the crystal lattice on macroscopic scale.

Lock-In as Scale-Invariant Principle

The lock-in mechanism (phase locking) is a geometric principle in FFGFT that acts on every scale. In the 4D torus, the condition reads:

$$\oint_{\mathcal{T}^4} \nabla_\mu \phi\, dx^\mu = 2\pi \cdot \mathcal{N}. \tag{22}$$

In the TFLN resonator — many orders of magnitude larger — the same formal condition applies:

$$\Delta\phi_{\text{cavity}} = 2\pi \cdot m \quad (m \in \mathbb{Z}). \tag{23}$$

The crystal lattice of lithium niobate spontaneously finds stable geometric configurations through phase separation and electro-optical coupling — it does not actively "search" but drifts naturally into the energetically most stable modes. In FFGFT language: the system converges to the preferred fixed points of the recursion $\Psi(t) = G(\Psi(t - T_{\text{rev}}), \xi)$, independent of which scale the system sits on.

Coherence as Consequence of Structural Fidelity

The high coherence time in TFLN waveguides is, in this picture, not a technical curiosity but a direct consequence of structural agreement: a system locked into geometrically stable modes decoheres more slowly than one driven against its natural geometry.

$$\tau_{\text{coh}}^{\text{TFLN}} \sim \text{ms} \ldots \text{s} \gg \tau_{\text{coh}}^{\text{Qubit}} \sim \mu\text{s}. \tag{24}$$

Atomic qubits fight against thermal noise and mass inertia — they are not anchored in the natural geometry of their scale. TFLN modes are.

.8 Self-Organisation as FFGFT Principle

The Effect: Spontaneous Pattern Formation Without External Programming

In modern TFLN-based chip manufacturing an effect appears that goes beyond mere lithography: the material organises *spontaneously* into stable geometric patterns, without each detailed structure having to be externally prescribed. Temperature and applied voltage allow control over *which* pattern emerges — but not whether self-organisation takes place. It always does.

FFGFT Explanation: Fixed Points of the Recursion

In FFGFT, self-organisation is not a special case but the normal state. The recursion

$$\Psi(t) = G\big(\Psi(t - T_{\text{rev}}),\ \xi\big) \tag{25}$$

has stable fixed points corresponding to the $1/n^2$ torus spectrum:

$$\Psi_n^* : \quad E_n \propto \frac{1}{n^2}, \quad n \in \mathbb{N}. \tag{26}$$

Every system — whether particle, atom, crystal lattice, or field mode — drifts naturally into one of these fixed points. There is no force that enforces this; the geometry of the torus admits no other stable states.

Temperature and Voltage as Geometric Control Parameters

Temperature and voltage do not change the principle of self-organisation but shift the *energy landscape*: they determine which fixed point n is energetically preferred.

$$n^*(T, U) = \arg\min_n \left[E_n - \mu(T) \cdot n - \lambda(U) \cdot n^2\right], \tag{27}$$

where $\mu(T)$ describes the thermal contribution and $\lambda(U)$ the electrical contribution to the effective mode energy.

- **Temperature** shifts $\mu(T)$: higher temperature favours higher modes (larger n), low temperature stabilises the fundamental ($n = 1$).
- **Voltage** shifts $\lambda(U)$: the applied voltage "presses" the modes into prescribed paths — in FFGFT language, an external boundary condition on the allowed winding numbers of the torus.

Structural Statement

The TFLN crystal lattice is a macroscopic system that seeks the same fixed points as a particle in the sub-Planckian regime — not because the scales would be related, but because the geometric recursion $\Psi(t) = G(\Psi(t - T_{\text{rev}}), \xi)$ generates the same stable states on every scale. Self-organisation is the visible expression of this scale-independent geometry.

.9 Experimental Evidence 1: Programmable TFLN Processors

In December 2025, a team from NTT, Cornell University, and Stanford University demonstrated a photonic processor based on lithium niobate [**?**].

Technical Specifications

- $\approx 10\,000$ programmable spatial degrees of freedom;
- Planar waveguide with manipulable refractive index;
- 96 % test accuracy in vowel classification;
- 49-dimensional vector operations in a single optical pass.

Quantum Networks and Multiplexing

Assumpcao et al. (2024) [**?**] demonstrated a TFLN platform with:
- Coupling losses < 1 dB/facet;
- Switching contrast > 20 dB;
- Electro-optical modulators with > 50 GHz bandwidth.

The bandwidth of > 50 GHz shows that TFLN hardware executes matrix operations with very high precision and stability — the structural agreement with the FFGFT matrix transformation is independent of the fact that the involved frequencies lie many orders of magnitude below ξ.

.10 Experimental Evidence 2: B-Meson Anomalies at CERN

Current Status (April 2026)

The LHCb experiment has measured a significant discrepancy in electroweak penguin decays of B mesons [**?**, **?**]:
- Decay channel: $B \to K^* \mu^+ \mu^-$ (~ 1 per 10^6 B mesons);
- Deviation from the Standard Model: **4.0** (≙ probability 1:16 000);
- Independent confirmation by CMS (2025).

FFGFT Interpretation of the Anomaly

The measured deviation concerns the Wilson coefficient C_9 of the effective $b \to s\mu^+\mu^-$ Hamiltonian:

$$\Delta C_9 \approx -1.0, \quad C_9^{\text{SM}} \approx 4.07 \quad \Rightarrow \quad \frac{|\Delta C_9|}{C_9^{\text{SM}}} \approx 25\,\%. \tag{28}$$

Hinweis

Quantitative Restriction. A direct derivation of this $25\,\%$ deviation from FFGFT parameters is *not* available. Simple expressions such as ξ, ϕ^{-4} or $|\ln(m_b/m_P)| \cdot \xi$ yield values too small by factors 10–10^3. The quantitative connection is an **open question**.

The modification of the decay rate by the fractal matrix structure has qualitatively the form:

$$\Gamma(B \to K^* \mu^+ \mu^-) = \Gamma_{\text{SM}} \cdot \left(1 + \alpha_{\text{FFGFT}} \cdot \mathscr{K}(\theta)\right), \tag{29}$$

where α_{FFGFT} is to be derived from the Z3 topology and the $1/n^2$ torus spectrum — this derivation is outstanding.

Connection to Lepton Flavour Universality Violation

Arslan et al. (April 2026) [?] analyse the decay $\Lambda_b \to \Lambda_c \tau \bar{\nu}_\tau$:

- Combined deviation of 3.8σ in the $R_{\tau/(\mu,e)}(D^{(*)})$ ratios;
- Maximum sensitivity in $\mathcal{K}_{1c}$, $\mathcal{K}_{2ss}$, $\mathcal{K}_{2cc}$, $\mathcal{K}_{4s}$.

Leptons of different generations appear in FFGFT as different harmonics of the same spacetime fundamental structure:

$$\xi_{\text{Muon}} = \phi^{-3} \cdot \xi, \qquad \xi_{\text{Tau}} = \phi^{-5} \cdot \xi. \tag{30}$$

Table 1: Leptons as harmonics of the fundamental frequency ξ

Lepton	Harmonic	Numerical Factor	Ratio to Predecessor
Electron	$\phi^{-1} \cdot \xi$	$\approx 0.6180 \cdot \xi$	(fundamental)
Muon	$\phi^{-3} \cdot \xi$	$\approx 0.2361 \cdot \xi$	$\phi^{-2} \approx 0.382$
Tau	$\phi^{-5} \cdot \xi$	$\approx 0.0902 \cdot \xi$	$\phi^{-2} \approx 0.382$

.11 Theoretical Evidence 3: Sub-Planckian Cut-off Scales

Effective Field Theory and Gravitation

Aynbund and Kiselev (2025) [?] establish a connection between the Planck mass and a natural sub-Planckian cut-off scale:

$$\Lambda_{\text{QG}} = \alpha \cdot \frac{M_{\text{red,Planck}}}{4\pi} \sim 10^{16}\,\text{GeV}. \tag{31}$$

This lies far below the Planck scale of $\sim 10^{19}$ GeV.

OLQEM and Fibonacci Quasi-Periodicity

Souza (2025) [?] investigates sub-Planckian scales in the framework of the Optical Lambda Quantum Energy Model:

$$I_{\text{OLQEM}} = \frac{4\lambda I_0}{n(\lambda)\,x} \cdot D(y) \cdot I(y) \cdot e^{-\gamma t} \cdot \left(1 + \frac{\chi^{(3)} I_0}{c^3 \varepsilon_0}\right) \cdot S_{\text{Fib}}\!\left(\frac{r}{a}\right). \tag{32}$$

The Fibonacci modulation reads:

$$S_{\text{Fib}}\!\left(\frac{r}{a}\right) = \sum_{k=0}^{\infty} \frac{\cos\!\left(2\pi\phi^k r/a\right)}{\phi^k}. \tag{33}$$

Hierarchical scaling of the length scales:

$$\ell_k = \frac{L_P}{\phi^k}, \quad k \geq 1. \tag{34}$$

For $k = 5$:

$$\ell_5 = \frac{L_P}{\phi^5} \approx \frac{1.616 \times 10^{-35}\,\text{m}}{11.09} \approx 1.46 \times 10^{-36}\,\text{m}. \tag{35}$$

This scale corresponds to FFGFT via $\ell_5 = c \cdot t_0$:

$$c \cdot t_0 = 2.998 \times 10^8\,\frac{\text{m}}{\text{s}} \times 7.188 \times 10^{-48}\,\text{s} = 2.155 \times 10^{-39}\,\text{m}. \tag{36}$$

Hinweis

Numerical Note. $\ell_5 \approx 1.46 \times 10^{-36}$ m and $c \cdot t_0 \approx 2.16 \times 10^{-39}$ m do not agree (factor ≈ 680). The correspondence is qualitative (sub-Planckian order), not numerically exact — which is discussed in detail in Doc. 182 (Universal scale).

Synthesis of the Sub-Planckian Models

Table 2: Comparison of sub-Planckian models

Model	Scale	Mechanism	Observable
FFGFT	$t_0 = t_P/7500$	Lock-in in the 4D torus via Z3 symmetry	TFLN phase patterns, B-meson anomalies
OLQEM	$\ell_k = L_P/\phi^k$	Fibonacci quasi-periodicity in photonics	Photonic interference in nonlinear crystals
EFT (Aynbund)	$\Lambda_{\text{QG}} \sim 10^{16}$ GeV	Gauge coupling to reduced Planck mass	B-meson decays (LHCb, CMS)

.12 Synthetic Analogue Computer: The FFGFT-TFLN Paradigm

Architecture

A TFLN-based analogue computer realises the FFGFT postulates not by approximation to the fundamental scale ξ, but through *structural isomorphism*: the mathematical operations — matrix transformation, convolution, lock-in — are the same on the crystal scale as on the sub-Planckian scale. The scale itself is irrelevant for the validity of the structure.

Structural Isomorphism

FFGFT does not claim that TFLN operates at ξ. FFGFT claims that the mathematical structure of the matrix transformation $\mathcal{T} : \mathbb{R}^4 \to \text{Mat}(N, \mathbb{C})$ is the same on every scale — from particle through atom to crystal lattice. TFLN is a macroscopic system that makes this structure visible.

Testable Predictions

Vorhersage

Prediction 1 — TFLN Interferometry. At optical intensities $> 10^{21}$ W/m^2, discrete phase jumps following the Fibonacci hierarchy should appear:

$$\Delta\phi_{\text{discrete}} = 2\pi \cdot \phi^{-k} \quad (k \in \mathbb{N}).$$

Vorhersage

Prediction 2 — B-Meson Analysis. The angular observables $\mathcal{K}_{1c}$ and $\mathcal{K}_{2ss}$ from [?] should deviate from SM predictions with higher statistics ($> 5\sigma$):

$$\mathcal{K}_{1c}^{\text{FFGFT}} = \mathcal{K}_{1c}^{\text{SM}} \cdot (1 + \alpha_{\text{FFGFT}} \cdot F_{\text{Fibonacci}}).$$

Vorhersage

Prediction 3 — Sub-Planckian Resonances. In nonlinear photonic crystals, modulation frequencies at multiples of $\xi \cdot \phi^{-k}$ should be detectable.

.13 Discussion: From Simulation to Hardware Transduction

The implication of FFGFT can be formulated precisely: photonic analogue computers based on TFLN show that the matrix transformation $\mathcal{T} : \mathbb{R}^4 \to \text{Mat}(N, \mathbb{C})$ is not restricted to the sub-Planckian scale. The crystal lattice — many orders of magnitude larger than atoms or particles — obeys the same geometric rules. That is the FFGFT principle of fractal self-similarity in macroscopic realisation.

The remaining open question is not technological but theoretical: can the quantitative strength of the fractal correction (e. g. in the B-meson anomalies) be derived from the recursion structure $\Psi(t) = G(\Psi(t - T_{\text{rev}}), \xi)$? This is treated in Doc. 183–184 for the RSA case; for hadronic decay processes, it is outstanding.

.14 Conclusion

- Spacetime is based on a fundamental tick $\xi = 7500/t_P \approx 1.391 \times 10^{47}$ Hz.
- The 4D phase torus is **not** to be identified with a 3D torus plus time continuum; time is integrated into the torus topology as a fourth geometric dimension.
- Fractal feedback enforces a lock-in of the fundamental frequency that stabilises matter.
- The 4D vector space is translated by the matrix translation $\mathcal{T}$ into a multidimensional matrix space.
- Matrix multiplication corresponds physically to a 4D torus convolution.
- TFLN photonics achieves the required bandwidth and programmable complexity.
- LHCb data show $\sim 4\sigma$ deviations from the SM — predictive for FFGFT; falsifiability at 5σ (DUNE 2028).

Kernpunkt

The coming years will show whether at 5σ a new understanding of spacetime hardware begins.

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